

# Keychain Kreatures — one-board fault report

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Complete one copy per board. Photograph before repair. Do not attach a real battery just to fill in this form. Mark unavailable tests BLOCKED or NOT RUN.

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## Identification

- Board: MAIN C.6 / POWER P.4 (circle one).
- Serial, supplier, order and batch/lot: \_\_\_\_\_
- Date received / tester / instructor: \_\_\_\_\_
- Bare PCB, partly assembled or fully assembled: \_\_\_\_\_
- Exact release ZIP name and SHA-256: \_\_\_\_\_
- Factory-approved BOM substitutions: \_\_\_\_\_
- Main: ESP32, display and SD-reader markings: \_\_\_\_\_
- Main: diagnostic firmware version/hash and known-good evidence: \_\_\_\_\_

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## First failure

- What was expected; what actually happened? \_\_\_\_\_
- First failed guide step and preceding passing step: \_\_\_\_\_
- Repeatable or intermittent? \_\_\_\_\_
- Did another unit pass the identical setup? Its serial: \_\_\_\_\_
- Smoke, smell, abnormal heat or current-limit event? \_\_\_\_\_
- Was a real cell connected? Approved part and condition: \_\_\_\_\_
- Any rework, reflash/erasure, jumper, substitution or forced connector? \_\_\_\_\_

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## Setup — attach a labeled photograph/diagram

- Source model, voltage, current limit and isolation: \_\_\_\_\_
- Attached cables, loads, modules and instruments: \_\_\_\_\_
- Where does every instrument ground/earth connect? \_\_\_\_\_
- Power: how is BAT\_NEG kept free of an external bypass to GND? \_\_\_\_\_
- Ambient temperature and temperature-measurement method: \_\_\_\_\_
- Meter/scope model, ranges, probe attenuation and test state: \_\_\_\_\_

Write both probe points. A number without its operating state or reference is not enough.

| Test /<br>signal | Ref / reference<br>points | Source, switch, load, firmware<br>state | Expected<br>context | Reading /<br>units |
|------------------|---------------------------|-----------------------------------------|---------------------|--------------------|
|------------------|---------------------------|-----------------------------------------|---------------------|--------------------|

Attach the completed checklist and measurement CSV. Scope captures must show channel scales, timebase, trigger, probe setup and board identity. Do not include Wi-Fi passwords or unrelated personal data.

## Requested factory response

- Compare the first failing stage with the exact schematic, BOM and Gerbers.
- Supply bare-board electrical-test results, assembly/hidden-joint inspection images and all substitutions for the lot.
- Power: confirm the approved via-in-pad process and WCSP/QFN assembly details.
- Identify the cause before proposing a part swap or undocumented jumper.
- Advise whether to preserve the unit untouched for return; obtain approval before rework.
- Provide corrective action, affected lot scope and post-repair test results.

## Repair and retest

| Date / person | Confirmed cause | Approved repair / part and lot | Retested steps | Outcome |
|---------------|-----------------|--------------------------------|----------------|---------|
|---------------|-----------------|--------------------------------|----------------|---------|

Final status: PASS AT RECORDED CONDITIONS / FAIL / BLOCKED / NOT RUN.

Instructor signature/date: \_\_\_\_\_

Passing is not production, battery-safety, USB, EMC or toy certification. Coordinate damaged-battery handling/return with the responsible lab and supplier; do not put it in an ordinary PCB return parcel.